

Modeling Radioresistance and Radiotropic Fitness in Multispecies Biofilms

A Computational Framework, Implementation Audit,
and One-Way Radiation-Transport Foundation

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Abstract

We present a computational framework and implementation audit for multispecies biofilm models under heterogeneous ionising-radiation fields. A proposed partial stochastic differential equation (PSDE) organises diffusion, directed motility, nutrient, radiation, melanin, interaction and forcing terms, while four executed simulations—two Langevin models, a Cellular Potts Model (CPM), and a radial radiodialysis solver—realise only subsets of that formalism. The CPM represents seven named species as computational biomass parcels rather than individual organisms. In the shipped parameterisation, the melanin term changes Metropolis acceptance by approximately 15.5%, whereas the direct per-species radiation term changes it by approximately one part in 10^5 for the two negatively signed species; only the product of melanin production and melanin coupling reaches the dynamics and is identifiable. A single-seed radial pattern is opposite to the direction implied by the direct tropism coefficient and is not clearly separated from the approximate seeding null with six parcels per species. A synthetic, explicitly non-calibrating CPM–OpenMC workflow executes one-way photon transport, constructive-solid-geometry mass normalisation, dose conversion, coordinate transfer, and cell-, lineage- and generation-level attribution, but no target physical parameter is inferred. The implemented biological phenotype is therefore radiotropism, not radiotrophy or radioresistance: the current CPM has neither biomass growth nor a survival process. The work establishes a reproducible computational foundation and identifies the measurements, uncertainty controls, and model revisions required before physical or biological feedback claims can be evaluated.

Keywords: biofilm modelling; Cellular Potts Model; ionising radiation; radiotropism; radioresistance; OpenMC; uncertainty quantification

1. Introduction

Microbial communities possess an extraordinary ability to adapt and survive under extreme environmental stressors, such as elevated radiation levels. Within such communities, melanized fungi and other extremophilic microorganisms have evolved specialized mechanisms that allow

them to persist in high-radiation environments. This study aims to develop a comprehensive mathematical framework that captures the complex interplay between these organisms’ nonlinear interactions, radiation sensitivities, and chemotactic behaviors within biofilm communities. By leveraging a combined approach that integrates Langevin dynamics, reaction-diffusion modeling, and a Hamiltonian-based formulation, we simulate and explore the spatiotemporal evolution of these communities under radiation stress and other external influences.

The practical motivation for this work lies in nuclear bioremediation. Contaminated sites such as the Chernobyl Exclusion Zone and the Hanford Nuclear Reservation host complex microbial communities in which melanized and radioresistant species coexist with metal-reducing bacteria capable of immobilizing radionuclides. Mathematical models that predict community dynamics under spatially heterogeneous radiation fields are essential for designing and optimizing bioremediation strategies. The framework developed here provides a testable computational scaffold for such predictions; it is not itself a calibrated predictor.

Five phenomena travel under one loose vocabulary and they are not interchangeable: *radiotropic* (directional growth or redistribution relative to a source), *radiotrophic* (radiation measurably enhancing growth or metabolism), *radioresistant* (elevated survival after exposure), *melanized radioprotective* (melanin measurably altering damage, survival or shielding), and *radiation-responsive* (any molecular, morphological or material change after exposure). A survival curve cannot calibrate a directional term, and a transcriptome cannot set a material density. Throughout this paper we use each word only for the endpoint it names. The model presented here expresses the first of the five, and the paper’s title says so.

2. Related Work

2.1 Melanized Fungi, Radioresistance, and the Contested Radiotrophy Claim

The observation that melanized fungi colonise high-radiation environments prompted a proposal that melanin transduces ionizing radiation into metabolically usable energy. That proposal remains contested, and this subsection reports it as a live dispute rather than as settled background. Zhdanova et al. [7] first documented the prevalence of darkly pigmented fungi within the damaged reactor at the Chernobyl Nuclear Power Plant, noting that species such as *Cladosporium sphaerospermum* dominated the mycobiota of the most contaminated structures. Subsequent work by Dadachova et al. [8] demonstrated that ionizing radiation alters the electronic properties of melanin, enhancing electron-transfer activity in melanized cells of *Cryptococcus neoformans* and *Wangiella dermatitidis*. Melanized cells exposed to radiation levels approximately 500 times above background grew significantly faster, accumulated more biomass, and incorporated three-fold more ^{14}C -acetate than non-irradiated melanized cells or irradiated albino mutants. Dadachova and Casadevall [9] subsequently proposed the term “radiosynthesis” to describe this phenomenon, drawing an analogy to photosynthesis in which melanin serves a role loosely comparable to chlorophyll. Three features of the founding experiment are usually omitted when it is cited and are reported here: the cell arm was a ^{188}Re beta field at 0.05 mGy h^{-1} rather than a photon field; the growth effects are single-digit percentages in dry weight; and non-irradiated

non-melanized cells grew better than non-irradiated melanized cells in the same experiment, so the melanized/albino contrast carries a baseline asymmetry present before any radiation was applied [53].

Shunk et al. [10] cultivated *C. sphaerospermum* aboard the International Space Station for 26 days, reporting a growth ratio of 1.21 ± 0.37 -fold against the ground control and lower counts beneath the fungal biomass. Neither result is a demonstration of radiotrophy and neither should be quoted without its interval: the stated ratio does not exclude unity, the attenuation comparison reached $p = 0.069$ on a single dish with one sensor pair, no dosimetric data were obtained (the sensors were PIN photodiodes), melanin was never measured, and the words “radiotrophic” and “radiation-shield” were struck from the title between preprint and publication [53]. The commonly circulated “ $2.17 \pm 0.25\%$ ” attenuation figure appears in no version of that work.

The radiation resistance of *Deinococcus radiodurans* operates through fundamentally different mechanisms. Rather than harnessing radiation for metabolic benefit, *D. radiodurans* survives doses exceeding 12 kGy through a combination of efficient DNA double-strand break repair via extended synthesis-dependent strand annealing (ESDSA), multiple genome copies enabling homologous recombination, and a potent manganese-based antioxidant system that protects proteins from oxidative damage [11, 12]. Daly et al. [13] showed that small-molecule Mn^{2+} -metabolite complexes specifically protect the proteome against radiation-induced carbonylation, establishing that protein protection, rather than DNA repair capacity alone, governs extreme radioresistance. This is the best-evidenced radiation phenotype in the set of species modelled here, and it is a survival phenotype.

Aspergillus niger is melanized, but the located isogenic evidence does not support a substantial melanin-mediated survival advantage under the tested radiation modalities. Cortesão et al. [51] compared *A. niger* N402 with its pigmentation-deficient $\Delta fwnA$ mutant MA93.1 and reported LD_{90} values of 366 versus 353 Gy (X-ray), 506 versus 567 Gy (helium ions), and 112 versus 112 Gy (iron ions). In the same dataset, disruption of non-homologous end joining reduced LD_{90} to 57, 55 and 50 Gy, respectively, indicating a much larger contribution from DNA repair than from pigmentation. Comparable null results have been reported for *Exophiala dermatitidis* wild type and its non-melanized *pks1* mutant across several ionising-radiation modalities [53].

2.2 Mathematical Models of Biofilm Communities

The mathematical modeling of biofilm communities has a rich history beginning with the foundational one-dimensional multispecies model of Wanner and Gujer [14], which coupled reaction-diffusion equations for substrate transport with biomass conservation laws to predict biofilm thickness and spatial species distributions. Xavier et al. [15] extended this framework to multiple spatial dimensions, providing a deterministic continuum approach to heterogeneous biofilm development. Individual-based modeling approaches, exemplified by the iDynoMiCS platform of Lardon et al. [16], complement continuum models by resolving cell-level stochastic interactions. Cross-diffusion biofilm models have been analyzed by Sonner et al. [17], who established math-

ematical properties of volume-filling multispecies systems that exhibit porous-medium-type degeneracy. Our PSDE framework extends these precedents by incorporating radiation-dependent fitness terms and phase-locking dynamics that couple species interactions to external radiation fields, moving beyond the substrate-limited growth paradigms that dominate existing biofilm models.

2.3 Hamiltonian and Stochastic Approaches in Theoretical Ecology

The application of Hamiltonian mechanics to ecological systems remains comparatively underexplored, despite the natural conservation laws that govern energy flow in ecosystems. Symplectic integration methods, which preserve the geometric structure of Hamiltonian flows, have been advocated for ecological modeling by Diele et al. [18] to ensure that numerical solutions maintain essential qualitative dynamics. The Kuramoto model of coupled phase oscillators [19] provides a well-established framework for synchronization phenomena in biological systems, from neural networks to circadian rhythms. Acebrón et al. [20] reviewed the model as a paradigm for synchronization, noting its generalization to systems with time delays, frequency-weighted coupling, and heterogeneous interaction topologies. Our phase-locking kernel $\Gamma_s(t, \mathbf{x})$ draws directly on this tradition, extending it to multispecies microbial communities where species-level oscillatory dynamics in growth and resource acquisition couple through shared radiation and nutrient fields. The use of Langevin stochastic dynamics to describe microbial population fluctuations follows naturally from the Fokker-Planck formalism and has been applied to competing microbial populations in chemostat systems [21], where stochastic differential equations revealed long-term competitive outcomes differing from deterministic predictions.

2.4 Coupling Transport-Code Radiation Fields to Cell-Based Models

One-way transfer of a Monte Carlo transport code’s dose field into a discrete cell-based model is an established, validated method class, though not previously instantiated with OpenMC. Liu et al.’s RADCELL toolkit couples Geant4/Geant4-DNA to CompuCell3D, extracting tumour geometry from the Cellular Potts model and returning per-cell dose and DNA damage [1]. García García et al.’s TOPAS-Tissue couples TOPAS/TOPAS-nBio to CompuCell3D and validated the one-way dose transfer against a clonogenic survival assay, recovering literature-consistent linear-quadratic coefficients [2]. Cogno et al. coupled the BioDynaMo agent-based framework to TOPAS-nBio to model radiation-induced lung fibrosis, describing theirs as the first coupled agent-based/Monte-Carlo model for that specific biological endpoint [3]. A targeted search found no published coupling of OpenMC to a Cellular Potts Model or to any other cell-scale biological or agent-based model; every documented OpenMC coupling is to nuclear-engineering multiphysics rather than biology. The one-way CPM–OpenMC path in Section 6 therefore extends an established transport-code-to-cell-model paradigm to a new transport code and a new biological target, rather than introducing the paradigm itself.

2.5 Bioremediation Context

The practical motivation for modeling these communities lies in nuclear bioremediation. *Shewanella oneidensis* MR-1 is a model dissimilatory metal-reducing bacterium capable of reducing

soluble U(VI) to insoluble U(IV) via extracellular electron transfer through c-type cytochromes [22, 23]. *Ochrobactrum intermedium* AM7, isolated from soil near the Kakrapar Atomic Power Station in India, tolerates high concentrations of thorium and produces exopolysaccharides (EPS) that complex with Th(IV), offering a pathway toward bioremediation of actinide-contaminated environments [24]. The combination of melanized fungi that persist in the high-dose zone with metal-reducing bacteria that immobilize radionuclides suggests the possibility of engineered biofilm consortia for in situ remediation. The fitness landscape framework introduced by Wright [25] and formalized through the NK model of Kauffman [26] provides a conceptual foundation for understanding how epistatic interactions among species traits shape adaptation under radiation stress, a perspective that our Hamiltonian kNN decision tree operationalizes quantitatively.

2.6 Radiotrophy Is Not Established for Any Species Modelled Here

Because this paper models seven named species, we audited the public evidence for the radiation phenotype of each rather than importing the labels on reputation. The audit [53] searched eleven data repositories at the API level (Dryad, Zenodo, Figshare, EBI BioImage Archive, BioStudies, IDR, EMPIAR, NASA OSD, heIDATA, DepositOnce, MINDS@UW), the NCBI sequence and functional-genomics archives, and Europe PMC, and assigned every phenotype label from the endpoint actually measured. Its verdict on the phenotype in this paper’s former title is TARGETED_LAB_EXPERIMENT_REQUIRED, and its finding is unambiguous:

Radiotrophy is not established for any of the seven species modelled.

The claim rests on a single 2007 paper covering two of the seven, and against it stand a same-species report of no substantial melanin protection under gamma, a same-species negative melanin-pathway result at 1 and 3 kGy, a same-genus null growth result under a TLD-anchored ^{137}Cs field, an isogenic null in *A. niger* across three ionizing modalities (Section 2.1), and a formal energy-budget critique holding that the deposited energy is insufficient and the media carbon sufficient to account for the observed growth [52]. “Not demonstrated” is not “disproved”, and we state it as the former.

Radioresistance, by contrast, is well evidenced for several of the seven, and the D_{10} values in Table 2 are real measurements. Radiotropism has its own candidate — Zhdanova et al.’s 2004 report of directed hyphal growth toward a ^{109}Cd source, with *C. sphaerospermum* 3176 among the responders — and that record is blocked twice over: whether its dose gradient was measured or calculated is unverified, and directed hyphal elongation has no representation in a model whose agents are isotropic biomass parcels with no elongation or connectivity term. Directed-growth modelling of fungal hyphae along a sensed gradient is a mature capability elsewhere: the Neighbour-Sensing model [4] computes each hyphal tip’s growth vector from local fields and their gradients, and the Vesicle-Supply-Center / hyphoid tip-growth model [5] gives the underlying tip-elongation geometry; adapting either to a computed radiation dose field is the model revision this framework would need to represent tropism directly rather than through a parcel’s acceptance bias. The nearest existing radiation-fungus model [6] predicts dose-rate-dependent proliferation *magnitude* in irradiated *C. neoformans*, not directional growth; no published model drives hy-

phal tropism from a computed radiation dose field, which is a gap this framework shares with the wider literature rather than a shortfall unique to isotropic biomass parcels. Three further audit verdicts bound what public data can do for this model at all: the CPM spatial machinery can be exercised but not calibrated on public data (PUBLIC_DATA_CAN_VALIDATE_PIPELINE_ONLY); no public source pairs a calibrated hydrated volume with wet mass, dry mass and matched blanks (TARGET_WET_BULK_MEASUREMENT_REQUIRED); and no dataset anywhere pairs an exact modelled species with both a calibrated 3-D biomass field and a characterised radiation exposure (PUBLIC_COMPONENTS_FOUND_BUT_NOT_PAIRED). Every parameter in Table 2 is therefore a literature-anchored prior, and a literature prior is not a calibration.

3. Mathematical Framework

3.1 Symbols and Notation

Table 1: Summary of principal mathematical symbols.

Symbol	Description
$F_s(t, \mathbf{x})$	Fitness function of species s , evolving over time t and space $\mathbf{x}$
$H(p, q)$	Hamiltonian function defining total system energy
p, q	Phase-space variables (generalised momentum and position)
$\eta_s(t, \mathbf{x}) = \sigma_s \xi(t, \mathbf{x})$	Stochastic noise: σ_s intensity; ξ spatiotemporal white noise
D_s	Species-specific passive diffusion coefficient
μ_s	Species-specific directed-motility (chemotaxis) coefficient
$P_{sj}(t)$	Permutation matrix describing species- s to species- j transitions
$I_\gamma(t, \mathbf{x})$	Ionizing (gamma) radiation field
$N(t, \mathbf{x})$	Non-ionizing (UV) radiation field
k_B	Boltzmann constant
$S[q(t)]$	Action functional for species trajectories
∇	Spatial gradient operator
$\Gamma_s(t, \mathbf{x})$	Phase-locked kernel for species s
$\beta_{s,\text{ion}}$	Ionizing radiation coefficient. In the PSDE (Eq. 1) a fitness-loss rate; in the CPM (Section 5) a <i>tropism</i> coefficient whose sign sets the direction of drift along the dose gradient. Magnitudes anchored to published D_{10} radioresistance data (Table 2)
$\alpha_{s,\text{nir}}$	Non-ionizing radiation coupling coefficient
θ_s	Adaptive phase-locking coefficient
γ_s	Gamma-radiation coupling coefficient in the adaptive-feedback term (Eq. 9)
Δ_s	Phase-locking synchronization deviation

3.2 Partial Stochastic Differential Equation (PSDE) for Species Fitness

The fitness of each species s is governed by the following PSDE:

$$\begin{aligned}
\partial_t F_s = & \nabla \cdot (D_s \nabla F_s) - \nabla \cdot \left(\mu_s \sum_{j=1}^n P_{sj}(t) F_j \right) + R_s + \sigma_s \xi(t, \mathbf{x}) \\
& - \beta_{s,\text{ion}} I_\gamma F_s + \gamma_s \Delta_s - \alpha_{s,\text{nir}} N F_s + \theta_s H_s + C_s
\end{aligned} \tag{1}$$

where $F_s \equiv F_s(t, \mathbf{x})$, $I_\gamma \equiv I_\gamma(t, \mathbf{x})$, and $N \equiv N(t, \mathbf{x})$. The nine terms encode:

- **Diffusion:** passive spatial spreading at rate D_s .
- **Directed motility:** chemotaxis and radiation-gradient-directed movement controlled by μ_s and the transition matrix P_{sj} .
- **Deterministic reaction:** $R_s(t, \mathbf{x})$, nutrient-dependent growth (see Section 3.7).
- **Stochastic noise:** $\sigma_s \xi(t, \mathbf{x})$, spatiotemporal white noise scaled by species noise intensity σ_s .
- **Ionising damage:** $-\beta_{s,\text{ion}} I_\gamma(t, \mathbf{x}) F_s$, radiation-induced fitness reduction.
- **Phase-locking adjustment:** $\gamma_s \Delta_s$, adaptive synchronisation gain from melanin-mediated energy transduction.
- **Non-ionising effect:** $-\alpha_{s,\text{nir}} N(t, \mathbf{x}) F_s$, UV/heat-driven fitness coupling.
- **Hamiltonian interaction:** $\theta_s H_s(t, \mathbf{x})$, energy-conserving inter-species force from the Hamiltonian (Section 3.3).
- **External forcing:** $C_s(t, \mathbf{x})$, environmental control inputs.

This subsection is specification, not method. No program in this work integrates Eq. 1. There is no fitness field F_s in any source file. A case-insensitive search for the identifier `F_s` across this repository’s Julia and R simulation sources — `biofilms_potts.jl`, `biofilms_potts_jacc.jl`, the three R codes, and the supporting scripts `validate_serial.jl`, `export_checkpoint.jl` and `scoby_3d.jl` — returns no match at all. The word “fitness” itself appears in exactly two places in that code, both header-comment attributions of this manuscript’s own title (`biofilms_potts.jl` and `biofilms_radiodialysis.R`), never as an implemented quantity. What the three R codes advance is a set of particle positions under overdamped first-order Euler steps, and the CPM advances a lattice configuration by Metropolis acceptance — neither is a discretisation of a field equation in F_s .

Four of the nine terms are additionally unrepresentable rather than merely unimplemented. The deterministic reaction R_s is nutrient-dependent *growth*, and the CPM has none: `state.nutrient` is written each step but never read by `compute_delta_H`, and `divide_cell!` has no trigger. The phase-locking adjustment $\gamma_s \Delta_s$ depends on the H_{kNN} machinery of Section 3.8, which is specified and unexercised. The non-ionising term $\alpha_{s,\text{nir}} N$ and the external forcing C_s have no counterpart in any state variable. Only diffusion, directed motility, noise and the ionising term have anything in the implemented models that corresponds to them, and even those correspond by analogy rather than by discretisation — the CPM’s radiation term is an acceptance bias, not a fitness sink.

Eq. 1 is therefore the framework this work proposes, and the simulations reported in Section 6 are separate, simpler models that share its vocabulary. Presenting the two as one system would be the central error this manuscript exists to avoid.

3.3 Hamiltonian Framework

The total Hamiltonian for the multi-species biofilm is:

$$H = \sum_{i=1}^N \left[\frac{1}{2} \rho_i v_i^2 + U_i(\mathbf{x}_i) \right] + \sum_{i \neq j} V_{ij}(r_{ij}) + \sum_{k=1}^K W_k(t, \mathbf{x}) \quad (2)$$

where ρ_i is the effective biomass density of species i [cells $\mu\text{m}^{-3} \times$ effective mass analog], $U_i(\mathbf{x}_i)$ is the potential energy in the nutrient concentration field, $V_{ij}(r_{ij})$ is the pairwise interaction energy, and $W_k(t, \mathbf{x})$ captures external energy contributions from radiation exposure and stochastic effects. For mutualistic interactions:

$$V_{ij}^{\text{mutual}}(r_{ij}) = -\gamma \exp\left(-\frac{r_{ij}^2}{\sigma^2}\right) \quad (3)$$

3.4 Symplectic Integration

Symplectic integrators preserve the geometric structure of the Hamiltonian flow, ensuring that energy and momentum do not drift over extended simulation windows. Phase-space trajectories evolve via the phase-locked canonical equations:

$$\frac{dq}{dt} = \frac{\partial H}{\partial p} - \Gamma_s(t, \mathbf{x}) F_s(t, \mathbf{x}), \quad \frac{dp}{dt} = -\frac{\partial H}{\partial q} + \Gamma_s(t, \mathbf{x}) F_s(t, \mathbf{x}) \quad (4)$$

The leapfrog/Verlet scheme is the intended numerical treatment: (1) evaluate H_s from the current state; (2) adjust the phase-locking deviation Δ_s ; (3) advance via a symplectic step [39].

This subsection is specification, not method. No program in this work performs symplectic integration. Eq. 2 carries a kinetic term, but no simulation reported here holds an independently-integrated momentum state: the R models advance positions under overdamped first-order Euler steps and the CPM is a Metropolis Markov chain, neither of which has a trajectory whose symplectic form could be preserved. No identifier named `momentum` exists anywhere in this repository’s simulation sources. `biofilms_3d.R` does hold an array its own comment calls a “velocity state,” used to carry the reflection direction across a cylindrical wall boundary — but it is recomputed in full from the current position and forces at every step ($\dot{\mathbf{x}} = \mu \mathbf{F} + \boldsymbol{\xi}$, the defining property of the overdamped limit), never advanced by its own equation of motion, and so is not the persisted inertial degree of freedom a symplectic integrator requires; the word survives in a comment where the physics does not. Implementing genuine symplectic integration is listed in Section 7.4.

3.5 Radiation Field Models

Ionizing radiation (gamma decay).

$$I_\gamma(t, \mathbf{x}) = I_0 \exp(-\kappa \|\mathbf{x}\|) \quad (5)$$

where I_0 is the initial intensity at the source and κ is the attenuation coefficient scaled by local biofilm density.

Non-ionizing radiation (UV oscillatory field).

$$N(t, \mathbf{x}) = N_{\text{UV}} \cos(\omega t) \quad (6)$$

3.6 Melanin Reaction-Diffusion

The spatial production and diffusion of melanin by the melanized fungi is modelled as:

$$\frac{\partial M}{\partial t} = D_M \nabla^2 M + \alpha_M \cdot n_{\text{RF}}(\mathbf{x}, t) \cdot I_\gamma(t, \mathbf{x}) \quad (7)$$

where D_M is the melanin diffusion coefficient [$\mu\text{m}^2 \text{s}^{-1}$], α_M is the radiation-driven production rate [$\mu\text{g cell}^{-1} \text{Gy}^{-1}$], and n_{RF} is the local melanized-fungal cell density [$\text{cells } \mu\text{m}^{-3}$]. Each term carries units of [$\mu\text{g } \mu\text{m}^{-3} \text{s}^{-1}$].

The sign of α_M is a modelling assumption and the evidence runs against it. A same-species, same-modality measurement in *C. neoformans* found the melanin-synthesis genes *LAC1* and *LAC2* down-regulated after 1 and 3 kGy of gamma, and a two-species pigmentation assay found pigmentation increased under UV but decreased under gamma, both at $P < 0.001$ [53]. Equation 7 as written, with $\alpha_M > 0$, therefore encodes the opposite sign to the only same-modality data located. It is retained here so that the assumption is visible and testable, not because it is supported.

3.7 Nutrient Uptake with Adaptive Feedback

$$R_s(t, \mathbf{x}) = \alpha_s C_s(t, \mathbf{x}) \left(1 - \frac{F_s(t, \mathbf{x})}{K_s} \right) + A_s(t) C_s(t, \mathbf{x}) \quad (8)$$

$$A_s(t) = \gamma_s \phi_s(t) - \beta_{s,\text{ion}} I_\gamma(t) \quad (9)$$

where $\phi_s(t) = \cos(\omega_s t - \theta_s)$ is the phase-alignment function and $A_s(t)$ is the adaptive feedback driven by phase-locking and local radiation stress.

3.8 Hamiltonian kNN Decision Tree

To model species transition probabilities modulated by neighbourhood phase states:

$$H_{\text{kNN}} = \frac{1}{\sigma_n} \sum_{j=1}^n P_{sj}(t) F_j(t, \mathbf{x}) \Gamma_s(t, \mathbf{x}) \quad (10)$$

3.9 Radiation Damage and Resilience

$$D_{\text{rad},s}(t, \mathbf{x}) = -\beta_{s,\text{rad}} I_\gamma(t, \mathbf{x}) F_s(t, \mathbf{x}) \quad (11)$$

Species with stronger DNA repair or antioxidant protection exhibit lower $\beta_{s,\text{rad}}$ values, reflecting the Mn^{2+} -dependent proteome protection described by Daly et al. [13]. The parameterisation additionally assigns low $\beta_{s,\text{rad}}$ to the melanized species. That is a modelling assumption and the isogenic evidence does not support it: melanin knockouts show no survival difference under ionizing radiation in *A. niger* [51] or in three independent *E. dermatitidis* studies across four modalities [53]. Where melanin measurably mattered in an evolution experiment, it reduced mutation accumulation without altering survival — an effect on damage that must not be generalised into a survival term.

3.10 Biofilm Mechanical Properties

The EPS matrix is modelled as a Finite Extensible Nonlinear Elastic (FENE) entropic spring:

$$U(r) = -\frac{1}{2} k R^2 \ln\left(1 - \frac{r^2}{R^2}\right) \quad (12)$$

where k is the spring constant, R is the maximum extensibility, and r is the current end-to-end distance. Viscoelastic creep follows the Kelvin-Voigt model:

$$\sigma(t) = E \varepsilon(t) + \eta \frac{d\varepsilon}{dt} \quad (13)$$

with stress relaxation time $\tau = \eta/E$ governing biofilm restructuring dynamics.

3.11 Radiodialysis Membrane Transport

To model contaminant ingress through the bioreactor membrane under radiation-driven permeability change, we introduce a minimal three-equation system coupling mobile contaminant transport, biosorption/bioreduction, and membrane structural damage.

Radiodialysis is a term coined for this three-equation system and is not an established term in radiation chemistry, membrane science, or radiobiology; a targeted search finds no scientific usage of it outside this work. The established descriptors for the underlying phenomenon are *radiolysis* (of water, into reactive oxygen species) and the resulting *lipid peroxidation*, and the literature on radiation-driven change in membrane permeability itself uses the descriptive phrase “radiation-induced membrane permeability change” (or hyperpermeability) rather than any single coined term.

Mobile contaminant — cylindrical reaction-diffusion. The concentration $c(r, t)$ of a dissolved contaminant (e.g. aqueous U(VI) , Cs^+) in the cylindrical biofilm domain satisfies:

$$\frac{\partial c}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D_{\text{eff}} \frac{\partial c}{\partial r} \right) - (k_{\text{ads}} X + k_{\text{red}} X_{\text{red}}) c + k_{\text{des}} s \quad (14)$$

where D_{eff} is the effective diffusivity through the EPS matrix, k_{ads} and k_{red} are the biosorption and bioreduction rate constants, X and X_{red} are the total and metal-reducing biomass densities, and $s(r, t)$ is the immobilised contaminant concentration (sorbed or reduced phase). Zero-flux

symmetry is imposed at $r = 0$.

Immobile phase — biosorption and bioreduction.

$$\frac{\partial s}{\partial t} = (k_{\text{ads}}X + k_{\text{red}}X_{\text{red}})c - (k_{\text{des}} + k_{\text{loss}})s \quad (15)$$

where k_{loss} accounts for precipitation, co-precipitation, or export of the immobilised fraction [48].

Membrane damage and radiation-driven permeability. Radiation progressively degrades membrane structural integrity $m(t) \in [0, 1]$:

$$\frac{dm}{dt} = -k_{\text{dam}} \dot{D}(R) m, \quad m(0) = 1 \quad (16)$$

where $\dot{D}(R)$ is the absorbed dose rate at the membrane wall ($r = R$). The effective permeability P_{eff} evolves with cumulative dose $D_{\text{cum}}(t) = \dot{D}(R) \cdot t$:

$$P_{\text{eff}}(t) = P_0 \exp(\alpha_P D_{\text{cum}}(t)) \quad (17)$$

consistent with radiation-driven permeability enhancement observed in Nafion membranes under ionising flux [47].

Robin boundary condition at $r = R$. The membrane couples interior concentration gradient to the external reservoir c_{ext} :

$$-D_{\text{eff}} \left. \frac{\partial c}{\partial r} \right|_{r=R} = P_{\text{eff}}(t) (c(R, t) - c_{\text{ext}}) \quad (18)$$

This is analogous to Donnan dialysis through a charged membrane [49, 46]. In the present implementation, however, membrane integrity $m(t)$ and $P_{\text{eff}}(t)$ are imposed as independent functions and do not form a closed constitutive feedback law. Equations 16–17 should therefore be read as candidate model components whose physical linkage remains to be selected and calibrated.

4. Parameter Estimation

Biologically justified parameter ranges for the six primary modelled species, derived from published experimental data, are given in Table 2. Three declarations govern how that table should be read.

These are priors, not calibrations. No value in Table 2 was fitted to data from this system, and no optimisation was performed (Section 5). A literature prior is not a calibration and must not be reported as one.

The $\beta_{s,\text{ion}}$ column spends radioresistance evidence on a tropism coefficient. The “Biological Basis”

entries are D_{10} and LD_{10} survival measurements, which are genuine and strong. In the CPM the coefficient they set does not kill anything; it decides which way a biomass parcel drifts in the dose gradient. That substitution is legitimate only as a declared modelling choice, and this is the declaration.

Two entries are weakly supported and are flagged rather than silently promoted. The $\alpha_M = 0.03\text{--}0.10 \mu\text{g cell}^{-1} \text{ Gy}^{-1}$ row for *A. niger* has no located direct measurement; the isogenic radiation study quantifies no melanin. A per-cell basis is also incompatible with the present entity semantics, because a simulated cell ID denotes a computational biomass parcel rather than one organism. The biological basis for the $\beta_{s,\text{ion}}$ row is weaker than for species with measured survival curves and is retained only as a flagged literature prior.

Seven of the twenty-six rows are on a per-cell basis this model cannot admit, and the conversion that would re-base them does not exist. The three α_M rows are $\mu\text{g cell}^{-1} \text{ Gy}^{-1}$ and the four K_s rows are cells mm^{-3} . Both denominators are *organisms*. The simulated entity is a computational biomass parcel with no calibrated organism count — one `cell_id` bundles cells with their water and extracellular material. Its computational lineage metadata does not supply a cells-per-parcel conversion, so neither quantity can enter the model as written. Re-basing them requires a declared cells-per-parcel mapping, which is exactly the quantity the entity-semantics contract refuses to invent: it depends on the lattice pitch, which is unselected, and on a packing density, which is unmeasured. The rows are retained because they are the literature values a future calibration would start from, and they are marked here so that no reader mistakes a per-organism prior for a model parameter. They are literature priors on an organism basis, not parameters of this model.

Table 2: Literature-derived parameter priors for modelled species. Values were not fitted to the simulated system; organism-based quantities are not directly compatible with the computational biomass-parcel representation.

Parameter	Symbol	Species	Range	Units	Biological Basis	Ref
Diffusion coeff.	D_s	<i>C. neoformans</i>	0.01–0.10	$\mu\text{m}^2 \text{s}^{-1}$	Passive Brownian diffusion, $\sim 5 \mu\text{m}$ yeast	[27]
Diffusion coeff.	D_s	<i>D. radiodurans</i>	0.05–0.50	$\mu\text{m}^2 \text{s}^{-1}$	Small coccoid ($\sim 1.5 \mu\text{m}$); higher D from size	[11]
Diffusion coeff.	D_s	<i>B. subtilis</i>	0.10–1.00	$\mu\text{m}^2 \text{s}^{-1}$	Rod-shaped motile cells; active motility augments D	[28]
Diffusion coeff.	D_s	<i>C. sphaerospermum</i>	0.005–0.05	$\mu\text{m}^2 \text{s}^{-1}$	Filamentous hyphae; low effective diffusion	[10]
Diffusion coeff.	D_s	<i>A. niger</i>	0.005–0.05	$\mu\text{m}^2 \text{s}^{-1}$	Filamentous; comparable to <i>C. sphaerospermum</i>	[9]
Diffusion coeff.	D_s	<i>S. oneidensis</i>	0.10–0.80	$\mu\text{m}^2 \text{s}^{-1}$	Facultative anaerobe, rod-shaped, flagellated	[22]
Motility coeff.	μ_s	<i>C. neoformans</i>	0.00–0.05	$\mu\text{m s}^{-1}$	Non-motile yeast; residual drift from EPS flow	[8]
Motility coeff.	μ_s	<i>D. radiodurans</i>	0.00–0.01	$\mu\text{m s}^{-1}$	Non-motile coccus	[11]
Motility coeff.	μ_s	<i>B. subtilis</i>	15–45	$\mu\text{m s}^{-1}$	Flagellum-driven; $\sim 25 \mu\text{m s}^{-1}$ at 30°C	[28]
Motility coeff.	μ_s	<i>C. sphaerospermum</i>	0.00–0.005	$\mu\text{m s}^{-1}$	Hyphal extension; $0.1\text{--}5 \mu\text{m min}^{-1}$ tip growth	[10]
Motility coeff.	μ_s	<i>S. oneidensis</i>	10–40	$\mu\text{m s}^{-1}$	Polar flagellum; comparable to <i>E. coli</i>	[22]
Ion. rad. sens.	$\beta_{s,\text{ion}}$	<i>C. neoformans</i>	$10^{-5}\text{--}10^{-4}$	Gy^{-1}	Melanised; $\text{LD}_{10} \sim 2\text{--}5 \text{kGy}$	[8, 9]
Ion. rad. sens.	$\beta_{s,\text{ion}}$	<i>D. radiodurans</i>	$5 \times 10^{-6}\text{--}5 \times 10^{-5}$	Gy^{-1}	$D_{10} \sim 12 \text{kGy}$; most radioresistant organism known	[11, 12]
Ion. rad. sens.	$\beta_{s,\text{ion}}$	<i>B. subtilis</i>	$10^{-3}\text{--}5 \times 10^{-3}$	Gy^{-1}	Spore $D_{10} \sim 1\text{--}2 \text{kGy}$; vegetative $\sim 0.2\text{--}0.6 \text{kGy}$	[29]
Ion. rad. sens.	$\beta_{s,\text{ion}}$	<i>C. sphaerospermum</i>	$10^{-5}\text{--}10^{-4}$	Gy^{-1}	Melanised Chernobyl isolate; $\approx C. neoformans$	[7, 10]
Ion. rad. sens.	$\beta_{s,\text{ion}}$	<i>A. niger</i>	$5 \times 10^{-5}\text{--}5 \times 10^{-4}$	Gy^{-1}	Melanised; less resistant than Chernobyl isolates	[9]
Ion. rad. sens.	$\beta_{s,\text{ion}}$	<i>S. oneidensis</i>	$5 \times 10^{-2}\text{--}10^{-1}$	Gy^{-1}	$D_{10} \sim 0.07 \text{kGy}$; extremely radiosensitive	[12]
Melanin rate	α_M	<i>C. neoformans</i>	0.05–0.15	$\mu\text{g cell}^{-1} \text{Gy}^{-1}$	Substrate-dependent; enhanced by radiation	[8]
Melanin rate	α_M	<i>C. sphaerospermum</i>	0.08–0.20	$\mu\text{g cell}^{-1} \text{Gy}^{-1}$	Constitutively melanised; high production rate	[7, 10]
Melanin rate	α_M	<i>A. niger</i>	0.03–0.10	$\mu\text{g cell}^{-1} \text{Gy}^{-1}$	DHN-melanin pathway	[9]
Carrying cap.	K_s	<i>C. neoformans</i>	$10^4\text{--}10^6$	cells mm^{-3}	Yeast biofilm under nutrient-rich conditions	[8]
Carrying cap.	K_s	<i>D. radiodurans</i>	$10^5\text{--}10^7$	cells mm^{-3}	Dense tetrads; high packing efficiency	[11]
Carrying cap.	K_s	<i>B. subtilis</i>	$10^5\text{--}10^7$	cells mm^{-3}	Dense biofilm with EPS matrix	[28]
Carrying cap.	K_s	<i>S. oneidensis</i>	$10^5\text{--}10^7$	cells mm^{-3}	Planktonic and biofilm modes	[22]
Phase-lock freq.	ω_s	All species	0.01–1.0	rad hr^{-1}	Circadian/ultradian metabolic oscillation	[19, 20]
Noise intensity	σ_s	All species	0.001–0.05	—	Thermal and demographic stochasticity	[21]

5. Computational Methods

The repository contains four executed simulation programs: three in R and one in Julia. The PSDE in Section 3 is a proposed organising formalism and is not discretised by any of these programs. No optimisation, Sobol analysis, or Morris screening was performed.

Two-dimensional Langevin trajectories (`biofilms.R`). Seven species are advanced on the $[0, 1]^2$ domain by an explicit Euler–Maruyama step written in base R, with Gaussian increments drawn independently per axis and positions clamped to the unit square. Species interact through a pairwise displacement term summed over all other species. The gamma field enters as a constant scalar drift $-\beta_s I_0$ added identically to both axes; this program therefore contains no spatial radiation gradient. Clustering uses base-R `kmeans` with $k = 4$, evaluated every ten steps. The reported run uses 1000 steps at $dt = 0.1$ and $I_0 = 0.2$.

Three-dimensional cylindrical bioreactor (`biofilms_3d.R`). This Shiny application advances seven species inside a cylinder by an explicit Euler step with wall reflection and a Beer–Lambert field $I_\gamma(r) = I_0 \exp(-\kappa r)$. Nutrient enters radially from the outer wall. A boolean `radiotropic` flag sets inward attraction for flagged species and outward repulsion for the others. Because the program has neither growth nor death, the flag denotes a spatial tropism rather than radiotrophy or radioresistance.

Radiodialysis finite-volume solver (`biofilms_radiodialysis.R`). Equations 14–18 are discretised on a one-dimensional radial grid of $N_r = 40$ points by a finite-volume method of lines and integrated with the adaptive, stiff-capable LSODA implementation in `deSolve` [50]. The cylindrical-axis limit and the Robin wall condition are represented explicitly in the semi-discrete operator.

Cellular Potts Model (`biofilms_potts.jl`). The CPM core is implemented in Julia using standard-library numerical components; HDF5, JACC and AMDGPU support checkpoint exchange and the portability port, while CairoMakie is used for figure export. The lattice is cubic with a cylindrical biological domain. The reported coupled run uses $N = 40$, six computational biomass parcels per species, and 100 MCS. One MCS is N^3 attempted copy operations. The active Metropolis rule contains four terms, $H_{\text{CPM}} = H_{\text{adh}} + H_{\text{vol}} + H_{\text{rad}} + H_{\text{mel}}$, at $T_{\text{CPM}} = 5.0$. The code has a manually invoked division primitive for lifecycle and lineage testing, but no growth law or automatic division trigger; it likewise contains no radiation-dependent survival process.

Coupled CPM–radiodialysis diagnostic. The Julia file includes a forward-Euler port of the radial transport system with automatic substepping against an explicit diffusion bound. Every ten MCS, the current implementation reduces CPM occupancy to scalar inputs for the radial solver. Those quantities are site-occupancy fractions, not biomass concentrations in g cm^{-3} , and the radial profile is averaged before use. The coupled radiodialysis path is consequently retained

as a numerical diagnostic and is blocked from physical interpretation until the occupancy-to-mass mapping and membrane constitutive law are repaired.

One-way CPM–OpenMC dose-transfer foundation. A portable HDF5 transport snapshot carries CPM geometry-defining labels and reference fields to a Python/OpenMC fixed-source photon model. OpenMC tallies heating per source particle on a Cartesian mesh. Per-bin material masses are obtained from constructive-solid-geometry material-volume estimates rather than full-bin approximations; heating is then converted to Gy per source particle and, when a source rate is supplied, to Gy s^{-1} . The physical dose field is transformed back to the Julia axis convention and joined to current cell, lineage and generation labels. The executed end-to-end validation uses an explicitly synthetic reference system with `target_calibration=false` and stops before changing the Hamiltonian, melanin drive, membrane state, accumulated dose, or CPM clock.

Reproducibility and implementation contracts. The serial CPM has a byte-pinned regression fixture. The JACC port is tested by per-kernel agreement on identical inputs rather than pathwise trajectory identity, because its checkerboard sweep and RNG ordering differ from the serial implementation. Python tests separately cover exchange orientation, config staging, material closure, exact-volume mass normalisation, dose conversion, label attribution, and fail-closed feedback-gate logic.

6. Results

6.1 Scope of the Executed Auxiliary Analyses

The two-dimensional Langevin program produces grouping under pairwise coupling, stochastic motion, and species-specific diagonal drift. Because its radiation term is spatially constant, it cannot support a claim about movement along a dose gradient. The three-dimensional Shiny program does contain a radial field and an explicit inward/outward tropism switch, but still has no growth or survival process. The motility–diffusion–radiation scatter is a visual summary of Table 2, not an independent result. The phase-locking functional H_{kNN} remains specified but unexercised, and the thorium half-life enters only as the premise for treating the radiation field as static on the simulated timescale.

6.2 CPM Radiation-Linked Term Magnitudes

The CPM acceptance rule reveals a strong scale separation between its two radiation-derived terms. At the shipped $I_0 = 1.0$ and $T_{\text{CPM}} = 5.0$, their per-move energy changes and Metropolis biases are shown in Table 3.

For the two negatively signed species, the direct radiation term changes acceptance by about one part in 10^5 , whereas the melanin term changes it by 15.5%. Radiation can therefore influence the CPM indirectly through the chain

$$I_\gamma \longrightarrow \text{melanin production} \longrightarrow M \longrightarrow \Delta H_{\text{mel}}.$$

Table 3: Magnitude of the radiation-linked terms in the CPM acceptance rule at the shipped constants $I_0 = 1.0$ and $T_{\text{CPM}} = 5.0$.

Term	ΔH	Acceptance bias
ΔH_{rad} , negative $\beta_{\text{ion}} = -5 \times 10^{-5}$	-5.0×10^{-5}	1.000010
ΔH_{rad} , $\beta_{\text{ion}} = 7.5 \times 10^{-2}$	$+7.5 \times 10^{-2}$	0.985
ΔH_{mel} at $M = 1.44$	-0.720	1.155

Only the product of the melanin-production scale and the melanin-coupling scale reaches the dynamics, so the two are not separately identifiable from this output. This comparison ranks only the two radiation-derived terms: hand-specified adhesion differences remain larger than both.

6.3 Single-Seed Radial Diagnostic

The coupled CPM run uses a 40^3 lattice, six parcels per species, 100 MCS, and seed 42. At the final snapshot, mean radial positions are approximately $r = 13.1$ for *C. neoformans*, $r = 11.5$ for *C. sphaerospermum*, and $r = 9.9$ for *B. subtilis*. The implemented radiation field is maximal on the axis and decays outward, so negative $\beta_{s,\text{ion}}$ biases the two fungi inward. Their observed ordering is in the opposite direction.

The initial parcel centres are sampled uniformly inside $r < 16$ lattice units. The corresponding centre-only null has mean radius 10.7 and per-parcel standard deviation 3.8. With six parcels per species, the observed *C. neoformans*–*B. subtilis* separation is about 1.4 standard errors of the difference. Because occupied parcel geometry evolves after seeding, the exact null must ultimately be obtained by replicate simulations rather than by the centre-only approximation. Figure 1 is therefore a diagnostic from one realisation, not evidence of established stratification.

6.4 Melanin-Field Diagnostic

The dimensionless melanin field increases over the 100-MCS run for all three producer classes (Figure 2). *C. sphaerospermum* reaches the largest final field value, $M = 1.44$, followed by *C. neoformans* and *A. niger*. The ordering follows the hand-specified production scales; these field values have no calibration to a physical melanin mass or concentration.

6.5 Radiodialysis Numerical Diagnostic

The R and Julia radiodialysis implementations solve the same reaction–diffusion system with independent time integrators. In the reported Julia run, membrane integrity decreases to $m = 0.779$ and the imposed ratio P_{eff}/P_0 reaches $e \approx 2.72$ after 100 MCS (Figure 3). The latter is exact by construction from the chosen constants rather than a fitted response. No physical MCS-to-second mapping exists, the dose-rate input is a placeholder, and the radial coordinate is not calibrated to centimetres; these quantities are therefore dimensionless diagnostics.

At the final snapshot, the wall concentration is approximately $c(R, t) = 0.87 c_{\text{ext}}$, while the unweighted mean over radial nodes is about $0.024 c_{\text{ext}}$ (Figure 4). The low interior value indicates limited penetration from an initially contaminant-free interior, not depletion of a pre-existing

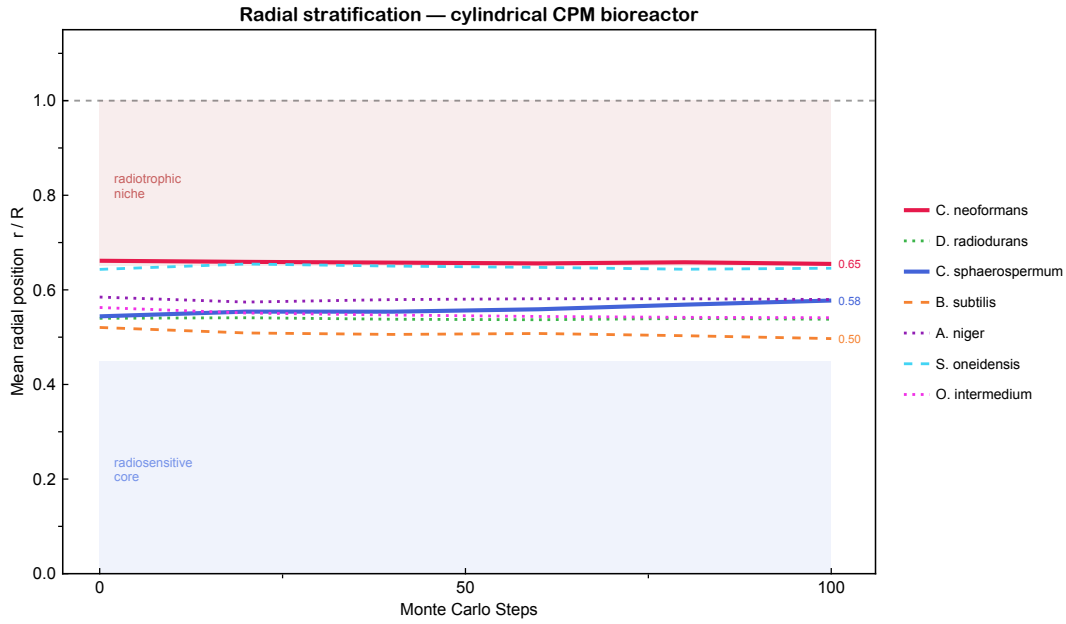


Figure 1: Mean radial position of the seven parcel classes over 100 MCS in the coupled CPM run. Shaded bands identify the inner high-radiation and outer lower-radiation regions of the imposed field; they do not assert biological niches. The displayed separation is a single-seed diagnostic and is not clearly separated from the approximate seeding null at the present parcel count.

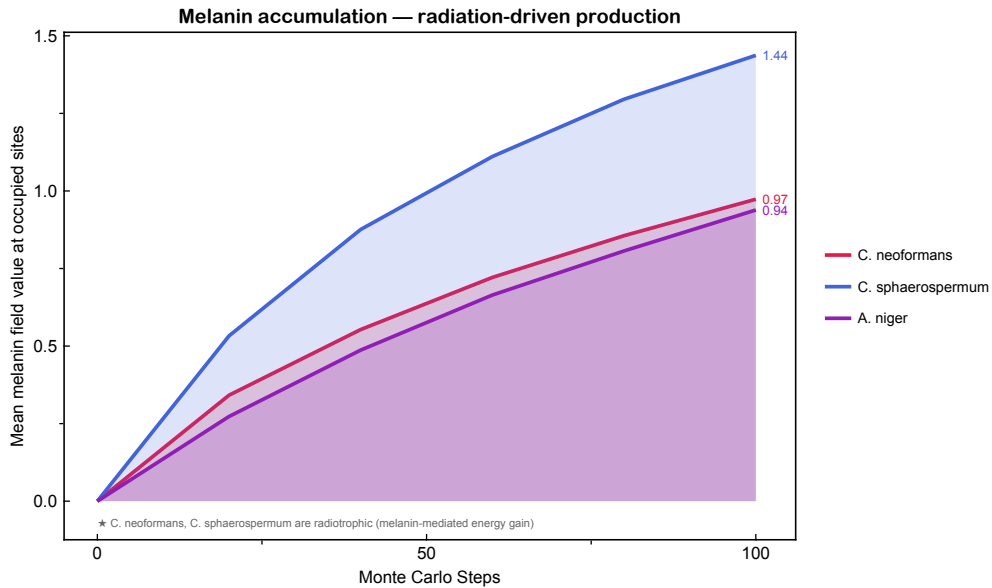


Figure 2: Mean melanin field at occupied sites for the three producer classes. Values are dimensionless model units; the sign and scale are uncalibrated and should not be interpreted as measured melanin mass or radiation-derived energy production.

inventory. Because the radial biomass profile is averaged before use and site occupancy is passed where the rate law expects mass concentration, this path remains blocked from physical interpretation.

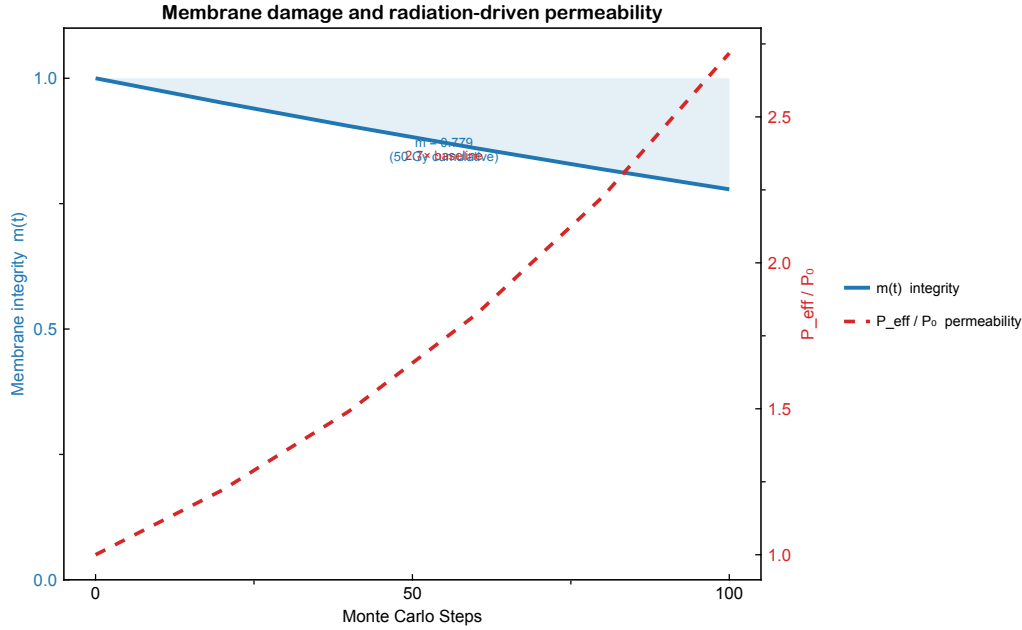


Figure 3: Independent imposed laws for membrane integrity $m(t)$ and normalised permeability $P_{\text{eff}}(t)/P_0$ over 100 MCS. The horizontal axis is not calibrated time, and the two curves are not a closed membrane-feedback law.

6.6 Synthetic One-Way CPM–OpenMC Integration

The one-way transport path executed end to end under an explicitly synthetic, non-calibrating reference system. A CPM snapshot was converted to an OpenMC biofilm-cylinder model; photon heating was normalised with constructive-solid-geometry material volumes; per-source dose was converted to a dose-rate field under a synthetic source rate; and the field was returned to Julia and attributed to cell, lineage and generation labels. Orientation probes and conservation checks passed across the exchange boundary. The run stopped before any physical dose changed the Hamiltonian, melanin, membrane state or CPM clock. This validates the implementation path, not the physical target.

7. Discussion

7.1 What the Implemented Models Establish

The implemented programs support a narrower conclusion than the full PSDE specification. The Langevin models demonstrate stochastic spatial grouping under chosen drift and interaction rules; they do not demonstrate radiation-enhanced growth or survival. The CPM term audit establishes that, within its radiation-derived pathway, the melanin term dominates the direct species-specific radiation term by several orders of magnitude. The single-seed radial output

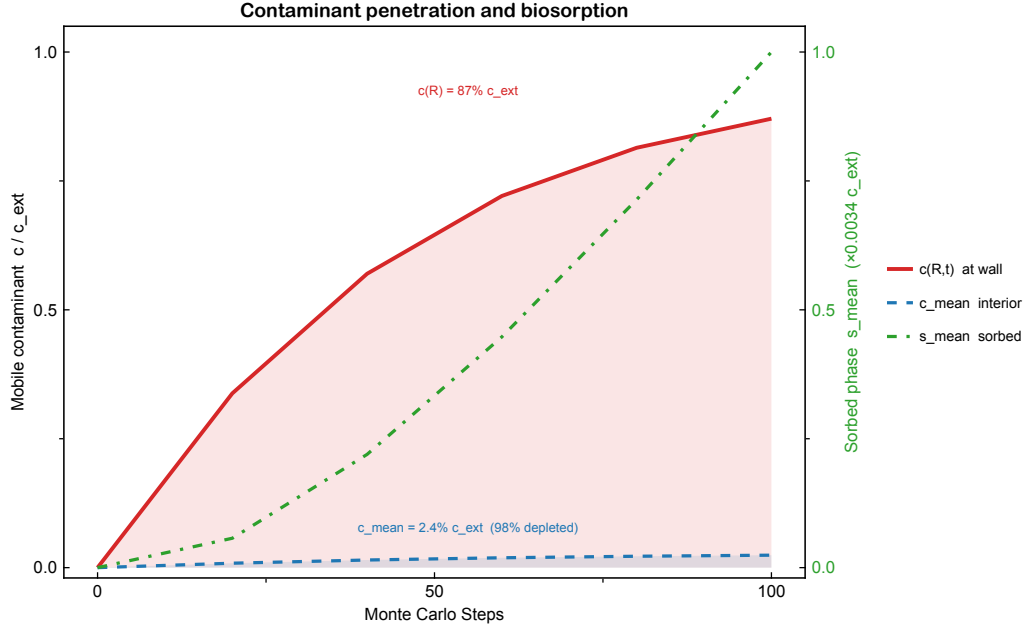


Figure 4: Dimensionless mobile and sorbed contaminant diagnostics over 100 MCS. The near-zero interior mean reflects limited penetration from the initially clean interior and must not be read as percentage removal. The sorbed phase uses its own display scale.

does not establish a radiotropic pattern, because its direction conflicts with the direct coefficient and its magnitude is not separated from the seeding null. The radiodialysis solvers establish a reproducible numerical solution of the stated semi-discrete equations, but the coupled Julia path does not yet possess a valid physical unit contract.

7.2 Phenotype Boundary

Radiotropism, radiotrophy and radioresistance refer to different observables. The present CPM can represent only directional redistribution in a spatial field. It cannot test radiotrophy because it has no biomass-growth process, and it cannot test radioresistance because it has no dose-dependent survival process. The published D_{10} and LD_{90} measurements in Table 2 remain biologically informative, but using them to set a tropism coefficient is a declared modelling analogy rather than a calibration of the phenotype they measured.

Melanisation is likewise not a universal proxy for either radiotrophy or survival. The literature contains organism-, genotype-, substrate- and radiation-modality-specific effects, including isogenic comparisons with little or no detectable pigmentation advantage. In the present CPM, melanin is best understood as an internal model field whose coupling dominates the direct radiation term; its physical mass, chemistry and radiobiological response remain uncalibrated.

7.3 Implications for Bioremediation and Transport Modelling

The one-way OpenMC foundation changes what can be tested next. Once a physically measured lattice pitch, hydrated density, water fraction, elemental composition, membrane geometry

and source model are supplied, the same snapshot-to-dose path can evaluate whether plausible material-state changes alter the local radiation field beyond transport and calibration uncertainty. That offline counterfactual question is logically prior to online biological feedback: if plausible material changes are transport-invisible, an expensive two-way coupling is not justified.

The current membrane equations do not yet support that loop. Membrane integrity and permeability are independent imposed laws, and the radiodialysis uptake term receives occupancy fractions where mass concentration is required. Fixed-membrane one-way dosimetry is still meaningful once the physical system is specified; dose-responsive membrane transport requires a separate constitutive and experimental programme.

7.4 Limitations and Future Directions

Structural model limits. Growth, survival, directed hyphal elongation, phase-locking, horizontal gene transfer, and viscoelastic mechanics are absent from the executed simulations. They cannot be recovered by fitting new coefficients to the present state variables. Adding baseline growth and death is the minimum model revision required before radiotrophy or radioresistance can be fitted. For directed hyphal elongation specifically, Section 2.6 names the adaptable machinery — a Neighbour-Sensing- or VSC-style tip-growth term [4, 5] — rather than an unbounded open problem.

Identifiability. Only the product of the melanin-production scale and the melanin-coupling scale reaches the CPM acceptance dynamics. Those parameters cannot be estimated separately from the same output without an independent measurement or reparameterisation.

Physical calibration. No value in Table 2 is fitted to the target system. Lattice pitch, wet bulk density, water fraction, closed hydrated elemental composition, seconds per MCS, source activity, and biological response normalisations remain unset. Public datasets can exercise parts of the acquisition pipeline, but no located source provides a paired target-compatible combination of calibrated three-dimensional morphology, wet and dry mass with matched blanks, elemental composition, and characterised radiation exposure [53].

Numerical resolution and uncertainty. The physical dose effect must be evaluated over independent transport replicates, declared mesh and ray-tracing convergence axes, correlated calibration draws, and explicit model-form scenarios. Numerical resolution is a bound to be reduced, not a random parameter to be averaged into the physical posterior. The scientific effect threshold must be declared prospectively before target feedback results are inspected.

Experimental next step. A target Reference D campaign should pair calibrated 3-D biomass imaging with matched wet/dry measurements, hydrated-volume support, abiotic blanks, closed wet-bulk composition, exact strain identities, and a characterised irradiation geometry. Culturing must begin only after the relevant facility has issued strain- and protocol-specific biosafety approval. The measured configuration can then support a biofilm-specific transport-resolution

study, a one-way physical dose calculation, and finally the scientific offline feedback gate.

8. Conclusion

This study separates a broad mathematical proposal from the smaller set of simulations that were actually executed. The active CPM represents seven species as computational biomass parcels and contains adhesion, volume, radiation and melanin terms. At the shipped constants, the melanin term is the dominant radiation-derived contribution to Metropolis acceptance, while the direct species-specific radiation term is negligible for the two negatively signed classes. The resulting single-seed radial pattern is diagnostic rather than established: it is opposite to the direct-term direction and is not separated from the seeding null at six parcels per species.

The implemented phenotype is radiotropism alone. Radiotrophy and radioresistance remain outside the current state space because the model has no biomass growth or dose-dependent survival. The radiodialysis solver is numerically defined but physically blocked by its occupancy-to-mass and membrane-law inconsistencies. In contrast, the synthetic one-way CPM-OpenMC path has executed through geometry export, photon transport, exact-volume dose normalisation, coordinate transfer and label attribution while remaining explicitly non-calibrating.

The immediate scientific requirement is therefore measurement rather than additional architectural coupling: paired three-dimensional morphology, wet and dry mass, water fraction, closed hydrated composition, source characterisation and a prospective effect threshold. Those inputs would permit a measured Reference D configuration, a biofilm-specific resolution study, and an offline test of whether plausible biological material changes are large enough to justify any online feedback.

Software and Data Availability

All source code, configuration schemas, tests, provenance ledgers, and figure-generation materials are available in the public repository at <https://github.com/aurascoper/Biofilms>. The supporting repository revision for this manuscript is `c7539a9`, the parent of the commit that adds this manuscript to the repository: a document cannot cite the commit that contains it, so it names the revision it was prepared against. No new biological specimens or experimental measurements were generated. Third-party public datasets are cited by their original records and are not redistributed as target calibration data.

Readers should note that the repository has continued past this revision, and that several numerical results published in it were subsequently corrected by the project’s own review: a false-positive control that could not fail was replaced, a squared difference reported as a variance was relabelled, an unprovenanced sensitivity table was withdrawn and re-measured, and one synthetic-gate verdict was withdrawn as not resolution-converged. None of those numbers appears in this manuscript, whose transport section reports the execution of a path rather than its results. The corrections are recorded in `data/claims_ledger.csv` and in `docs/calibration/`.

Ethics, Biosafety, and Regulatory Statement

This manuscript reports software development, computational simulation, code audit, and analysis of published or publicly available information. It involved no human participants, vertebrate animals, live microbial culture, recombinant or synthetic nucleic-acid experiment, or physical ionising-radiation experiment. No institutional human-subject, animal-use, biosafety, or radiation-use approval was required for the work reported here. Any future target culture or irradiation campaign will begin only after strain-, protocol-, facility- and source-specific institutional approvals have been issued; no such campaign or approval is claimed in this manuscript.

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